Bernoulli NML Selects Relative-Periodic Domains

A fitted relative-periodic scaffold isolates defect masks that retain visible structure in several 2D cellular automata

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Main result. A single Bernoulli-NML selector fits a global relative-periodic scaffold to finite spacetimes in 1D, 2D, and 3D cellular automata. In the strongest 2D cases, the remaining defect mask is sparse, but it does not collapse to featureless noise.

Problem

Cellular automata often mix broad regular domains with particles, interfaces, and other persistent defects. The practical finite-data question is which global period and shift best explain the observed spacetime.

Guiding idea. Fit the global scaffold first, then inspect the residual mask that remains.

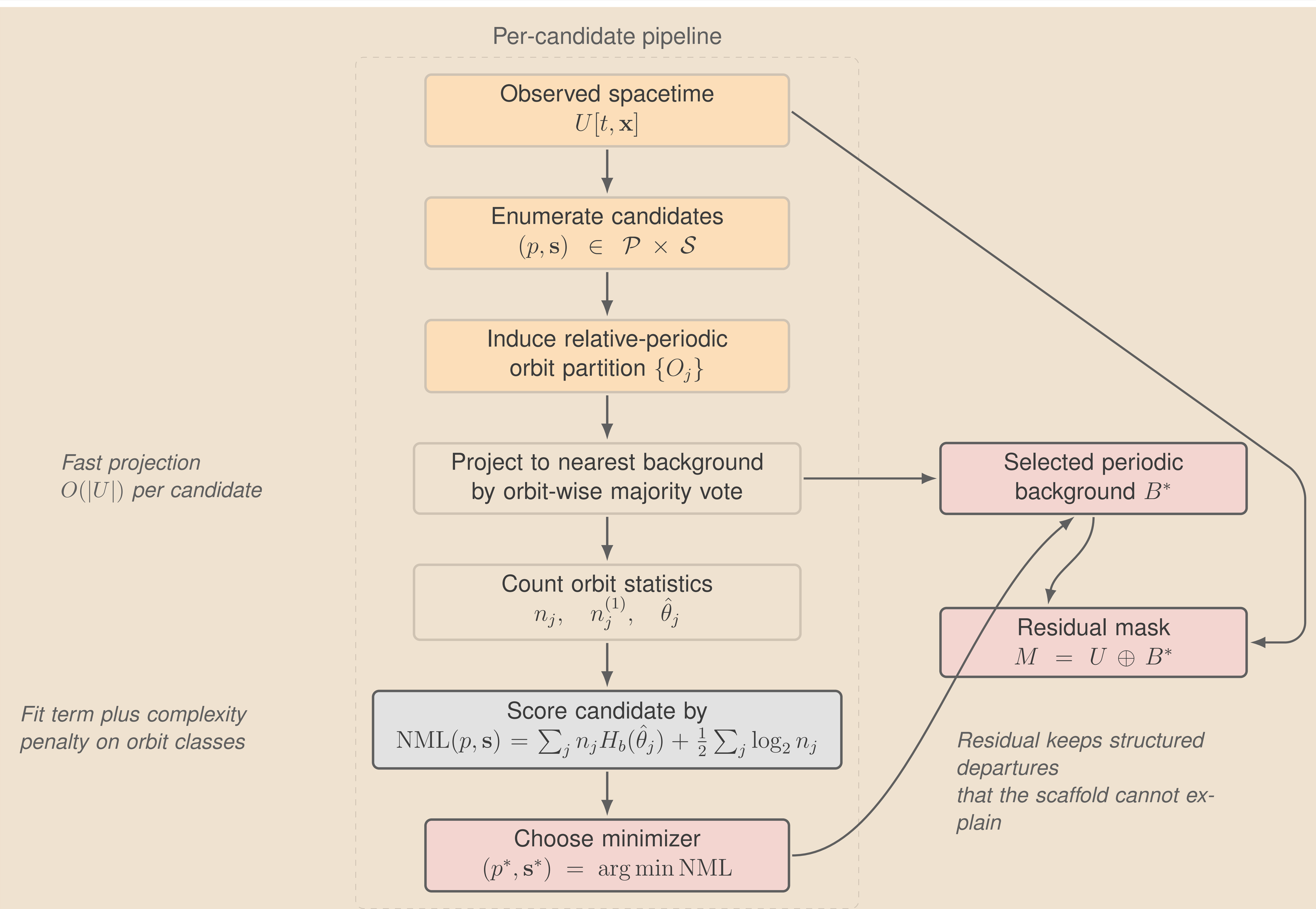
Selector

- A candidate $(p, \mathbf{s})$ defines a relative-periodic model class and its orbit partition.
- Majority vote on each orbit yields the fitted background $B^*(p, \mathbf{s})$.
- The residual mask is $M = U \oplus B^*(p, \mathbf{s})$, and Bernoulli NML ranks the candidates.

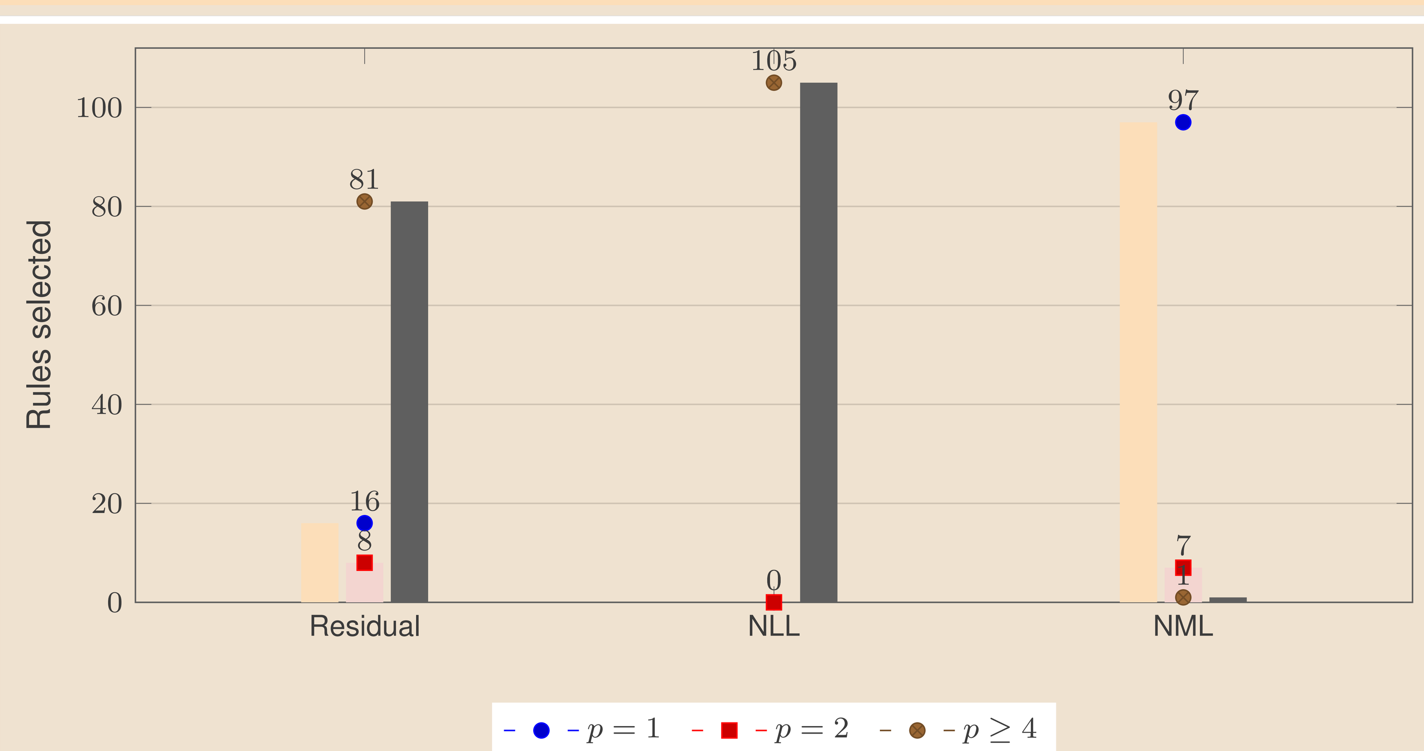
$$\mathcal{B}(p, \mathbf{s}) = \{B : B[t + p, \mathbf{x} + \mathbf{s} \bmod \mathbf{D}] = B[t, \mathbf{x}]\}$$

$$\mathrm{NML}(p, \mathbf{s}) = \sum_j n_j H_b(\hat{\theta}_j) \; + \; \frac{1}{2} \sum_j \log_2 n_j$$

Here n_j is the size of orbit class j and $\hat{\theta}_j$ is its fitted Bernoulli parameter. Pure orbit classes lower the fit term, but extra template bits raise the penalty.



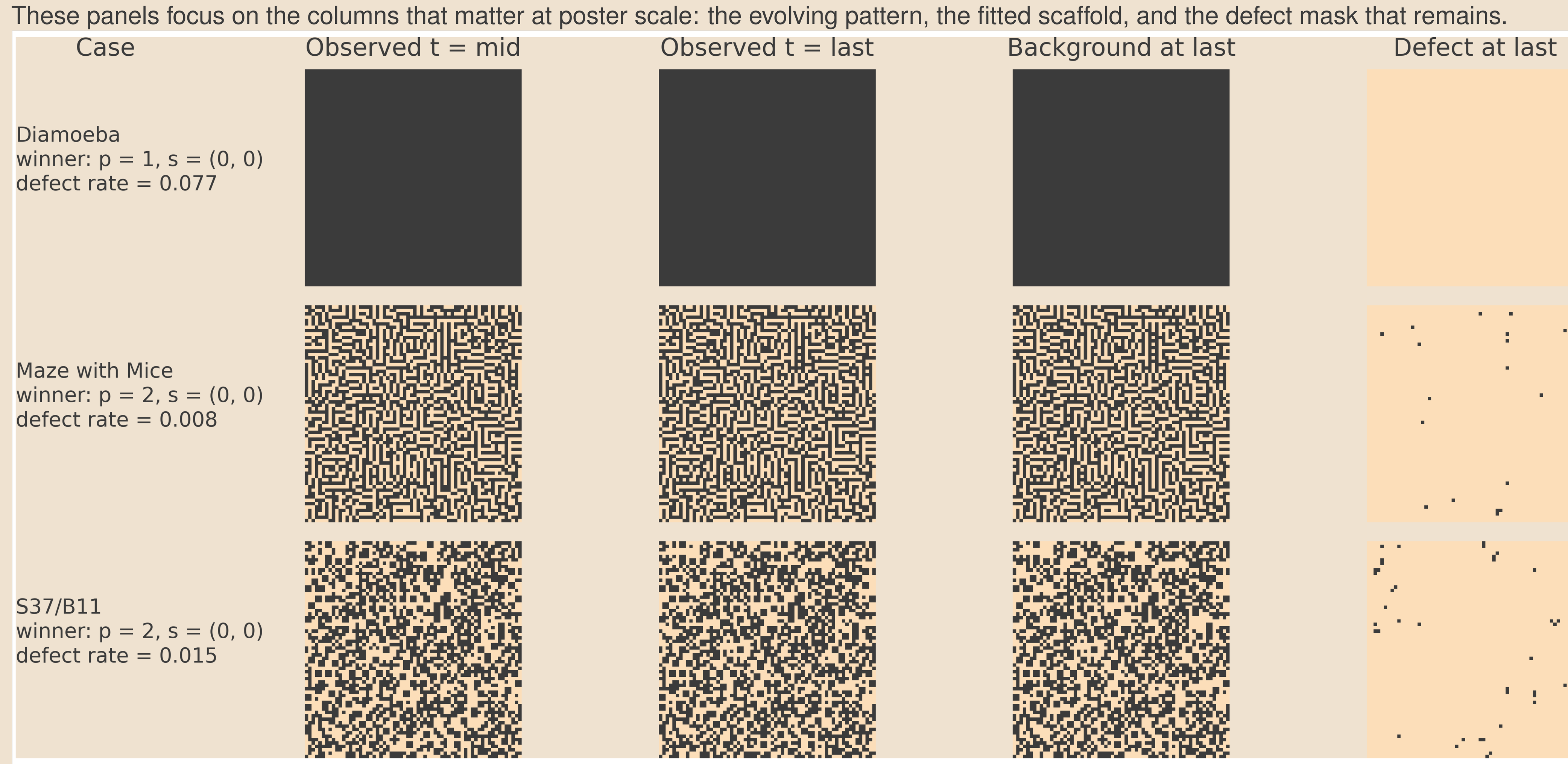
Selector Ablation



Residual minimization and Bernoulli NLL both inflate period because fit can only improve along refinement chains. The Bernoulli NML penalty is what keeps the selector parsimonious.

2D Main Result

These panels focus on the columns that matter at poster scale: the evolving pattern, the fitted scaffold, and the defect mask that remains.

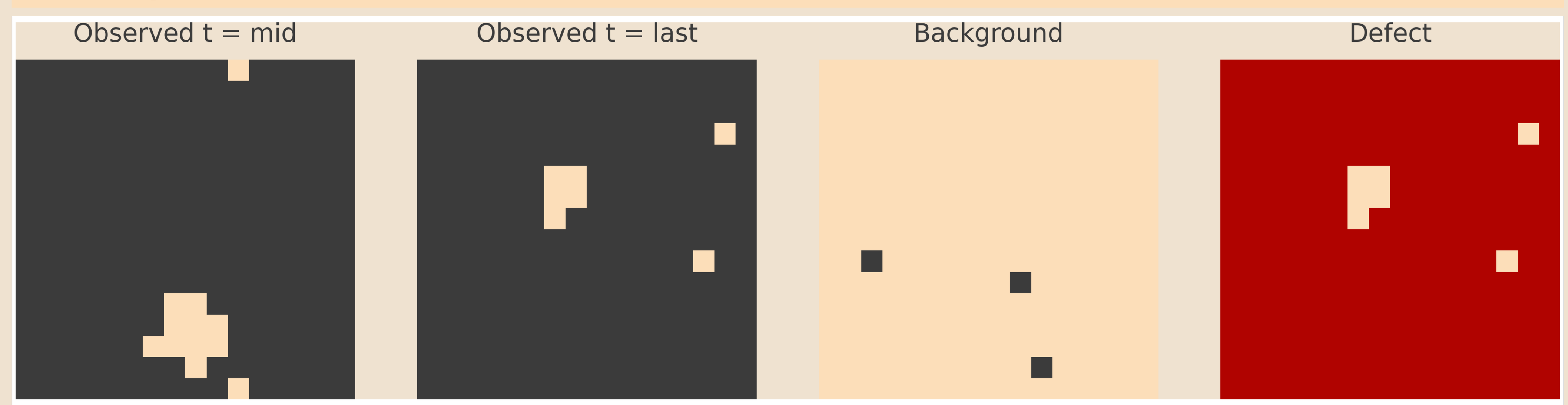


Main visual claim. In 2D, the fitted scaffold explains broad regular domains, but the residual mask does not become featureless. Instead, it often retains sparse coherent structure.

Focal 2D Cases

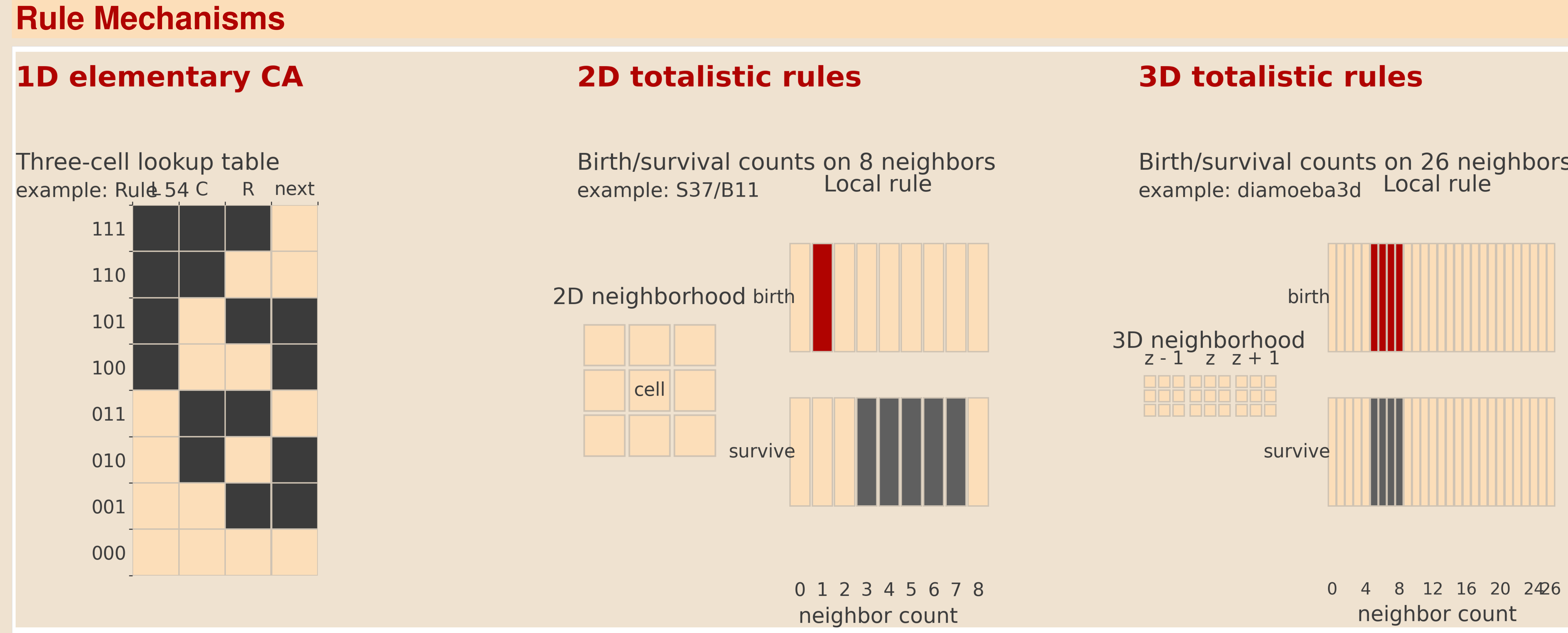
- In **Diamoeba**, large breathing domains lead the selector to choose $(p, \mathbf{s}) = (1, (0, 0))$, with defect rate 0.077.
- In **Maze with Mice**, a coherent maze background yields winner $(2, (0, 0))$, with defect rate 0.008.
- In **S37/B11**, a low-defect background coexists with persistent residual structure; the presentation run gives winner $(2, (0, 0))$ and defect rate 0.015.

3D Example



Diamoeba3d, max projection over z. Left to right: the active volume at mid horizon, the later observed state, the fitted background, and the residual interfaces.

Rule Mechanisms



Readable local-rule summary: 1D lookup tables, 2D Moore-neighborhood birth/survival counts, and 3D totalistic thresholds.

Stability Across Horizon

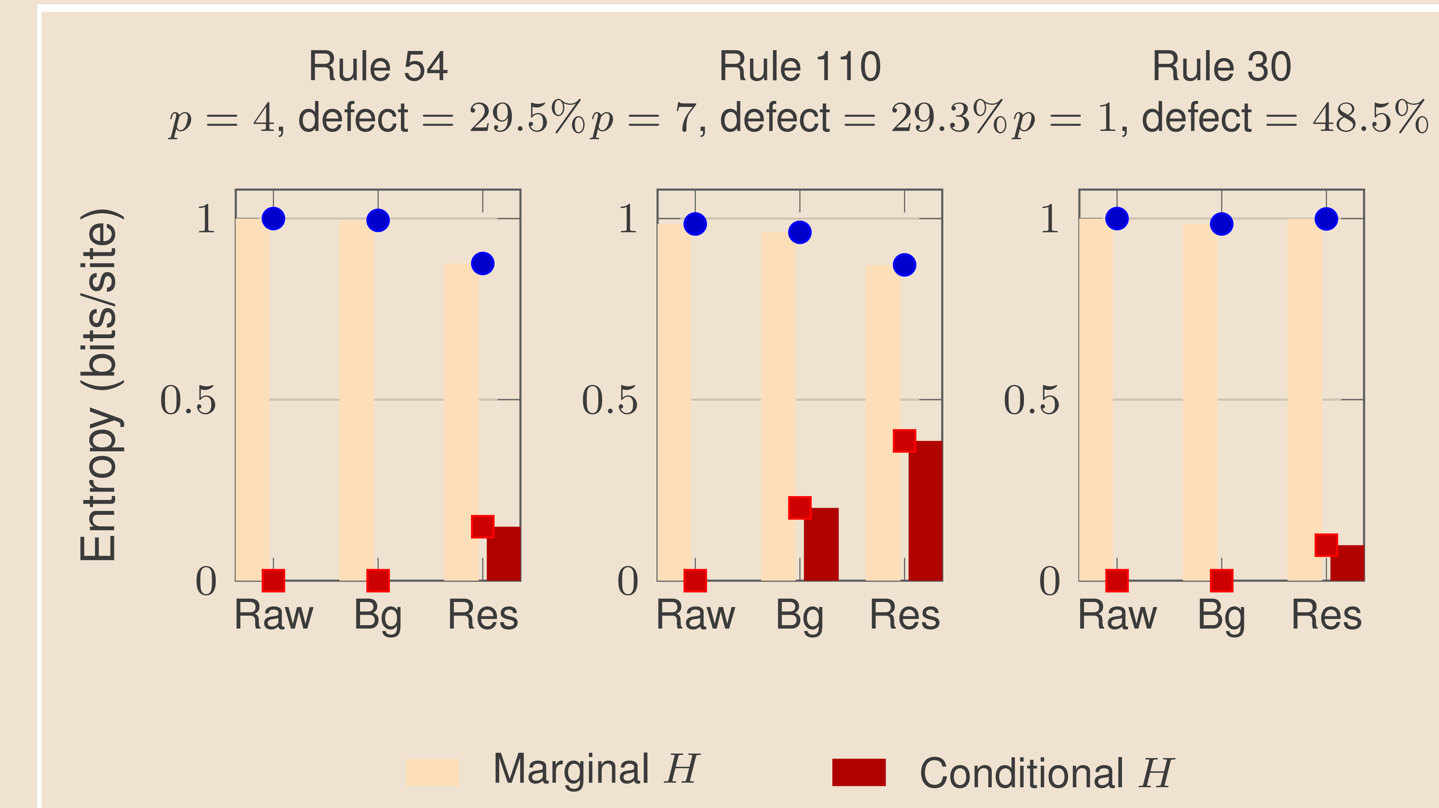
As the observation horizon grows, the selected period usually stabilizes quickly.

$$\mathrm{NML}(c, T) = \lambda_c N(T) + \frac{k_c}{2} \log_2 T + o(N(T))$$

When candidates have distinct asymptotic rates, the worse rate is eventually excluded and the winner locks.

- **ECA-54** locks immediately to period 4.
- **ECA-110** passes through a short transient and then locks to period 4 with shift -2.
- **S24/B11** changes later, because longer horizons reveal additional periodic structure.
- **Diamoeba3d** corrects a small-sample transient and then locks to period 1.

Residual Entropy



The manuscript entropy decomposition makes the poster's qualitative point more precise: the fitted periodic background keeps most predictability, while the residual has substantially lower conditional entropy than marginal entropy. The mask is sparse, but it is not white noise.

Takeaways

- One Bernoulli-NML selector provides usable periodic scaffolds in 1D, 2D, and 3D cellular automata.
- The complexity penalty changes the selected period qualitatively; it is not a cosmetic adjustment.
- In 2D, residual masks expose structured defects and therefore make broader surveys feasible.
- The selector is a front end for richer local analysis, not a replacement for computational mechanics.

Repository. `https://github.com/zitterbewegung/relative_symmetry_repair`

Figure sources. TikZ schematics live in `poster/figures/`. Poster-facing rule panels are regenerated from `scripts/alife_rule_diagrams.py`. The entropy comparison is adapted from the Complex Systems manuscript bundle.

Scan for repo, poster assets, and scripts.